NANO 2314 Quiz 08 — Review Packet

Practice for Quiz 08, Friday, October 30, 2026 · Fall 2026 · Prof./Coach Emori

17 problems on the ideas Quiz 08 covers, 15 of them free response: write your answer, with your work, in the space given. Try Part 1 first, then check yourself in Part 2. Multi-step answers are drawn step by step, and the **In words** box under each is the kind of written answer that earns full credit. No calculator needed.

Give every period and frequency its unit, with the right prefix: units cost points on every quiz. The **Quiz 08 Practice Drill** (https://claude.ai/artifact/HUwP4HpryHxUJoDTeMB3aC) gives fresh multiple-choice versions of all of these ideas, with new numbers each round.

Topic	Practices	Problems
1. Periods and frequencies, with units	T = 1/f with prefixes, the same time written other ways	1–3
2. Signal × reference on aligned graphs	sketching the three graphs, the product, labeling the time axis	4–6
3. Orthogonality	the integral of the product, which pairs are orthogonal	7–11
4. Equations from a diagram	moving rates to a system of equations, and back	12–15
5. Spreadsheet formulas	writing the formula with dollar signs, its value, copying it down	16–17

Part 1 · Problems

1. Periods and frequencies, with units

Problem 1 FREE RESPONSE Period ↔ frequency, with the prefix

Find each period: (a) $f = 40\ \text{kHz}$ (b) $f = 2\ \text{MHz}$ (c) $f = 250\ \text{Hz}$ (d) $f = 5\ \text{GHz}$

Find each frequency: (e) $T = 25\ \mu\text{s}$ (f) $T = 4\ \text{ms}$ (g) $T = 500\ \text{ns}$

Give each answer with a unit, in the prefix that keeps the number simple.

Problem 2 FREE RESPONSE The same time, written other ways

a. Write $25\ \mu\text{s}$ in milliseconds, and in seconds with a power of ten.

b. Write $2.5\ \text{MHz}$ in kilohertz, and in hertz with a power of ten.

Problem 3 FREE RESPONSE How many periods fit?

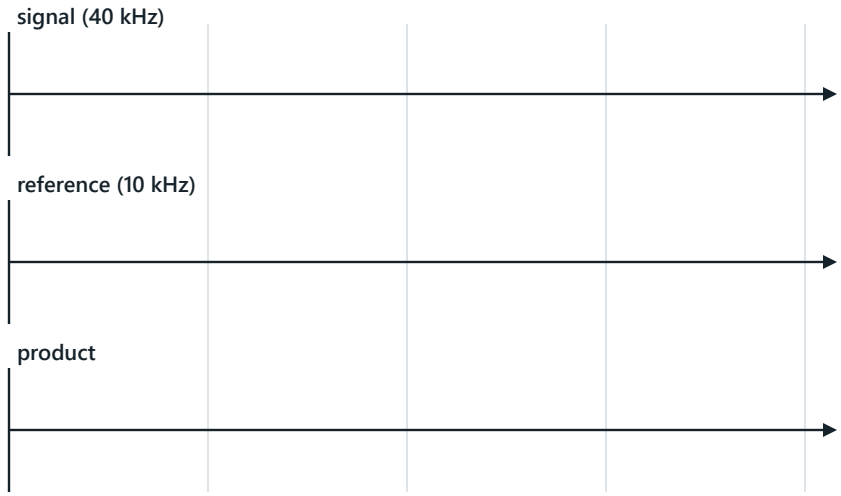
A $40\ \text{kHz}$ signal is multiplied by a $10\ \text{kHz}$ reference. (a) How many periods of the signal fit in one period of the reference?

b. A sketch shows the signal over 8 of its periods. How many periods of the reference does it show?

2. Signal × reference on aligned graphs

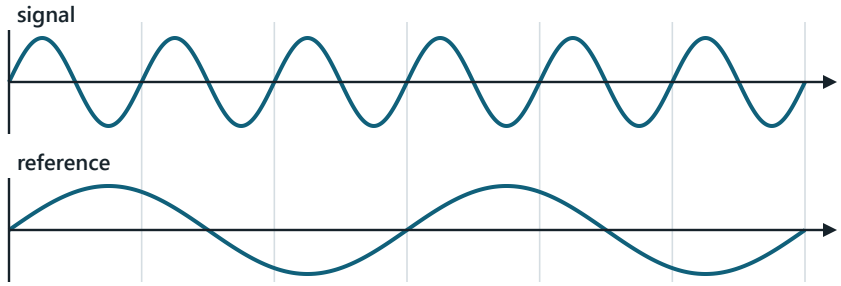
Problem 4 FREE RESPONSE Sketch signal, reference and product

A 40 kHz signal is multiplied by a 10 kHz reference; both are sine waves with the same amplitude. On the aligned axes below, sketch the signal over 4 of its periods (top), the reference over the same time (middle), and their product (bottom). Label the horizontal (time) axis with numbers and a unit.

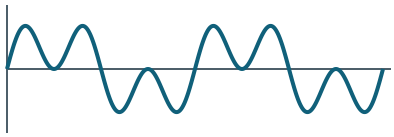


Problem 5 MULTIPLE CHOICE Pick the product

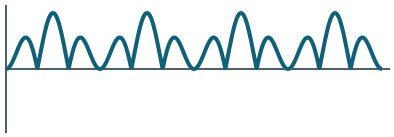
The top graph is a signal and the bottom one a reference, over the same time. Which curve is their product, signal × reference?



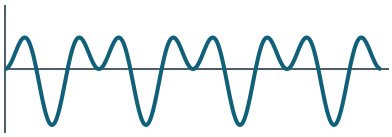
A.



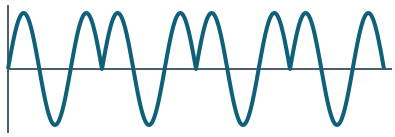
B.



C.

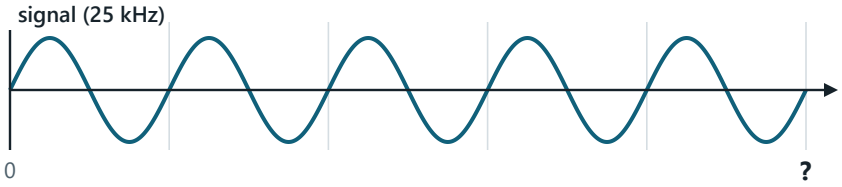


D.



Problem 6 FREE RESPONSE Label the time axis

This graph shows a $25\ \text{kHz}$ signal starting at $t = 0$. What time belongs at the right end of the axis (the "?")? Show your work.



3. Orthogonality

Problem 7 FREE RESPONSE What does orthogonal mean?

In your own words, what does it mean for two functions to be orthogonal? How could you check it from a graph of their product?

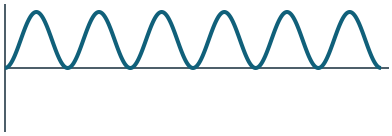
Problem 8 FREE RESPONSE The integral of the product in Problem 4

For the product you sketched in Problem 4, what is the integral from $t = 0$ to $t = 100\ \mu\text{s}$? How do you know? Are the $40\ \text{kHz}$ signal and the $10\ \text{kHz}$ reference orthogonal?

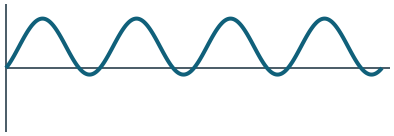
Problem 9 MULTIPLE CHOICE Which product adds up to zero?

Each curve is a signal × reference product over a window that holds whole cycles of both. Which product has zero total area (integral)?

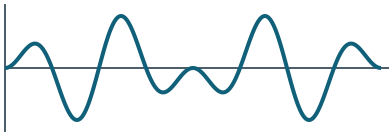
A.



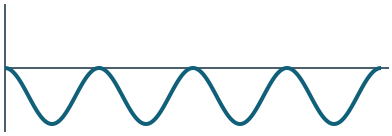
B.



C.



D.



Problem 10 FREE RESPONSE Which pairs are orthogonal?

Over $0 \le t \le 1$ s, which of these pairs are orthogonal? Explain briefly.

a. $\sin(2\pi \cdot 2t)$ and $\sin(2\pi \cdot 5t)$

b. $\sin(2\pi \cdot 3t)$ and $4\sin(2\pi \cdot 3t)$

c. $\cos(2\pi \cdot 1t)$ and $\cos(2\pi \cdot 4t)$

d. $\cos(2\pi \cdot 2t)$ and $-\cos(2\pi \cdot 2t)$

Problem 11 FREE RESPONSE The integral, with units

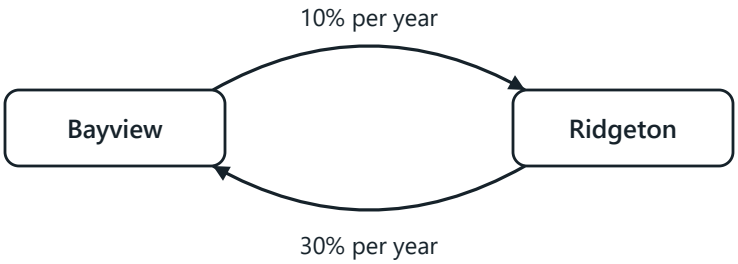
A signal $2\ \text{V}\sin(2\pi \cdot 10\ \text{kHz} \cdot t)$ is multiplied by a reference $5\ \text{V}\sin(2\pi \cdot 10\ \text{kHz} \cdot t)$. (a) What is the integral of the product from $t = 0$ to $T = 300\ \mu s$? Include the unit.

b. What would it be if the reference were $5\ \text{V}\sin(2\pi \cdot 30\ \text{kHz} \cdot t)$ instead?

4. Equations from a diagram

Problem 12 FREE RESPONSE Write the system from the diagram

Every year, people move between Bayview and Ridgeton as shown. Let b_n and r_n be the populations of Bayview and Ridgeton in year n. Write a system of two equations that gives b_n and r_n in terms of b_{n-1} and r_{n-1}.



Problem 13 FREE RESPONSE One year later

In year 0, Bayview has 600 people and Ridgeton has 400 (rates as in Problem 12). How many live in each place in year 1?

Problem 14 FREE RESPONSE What each term means

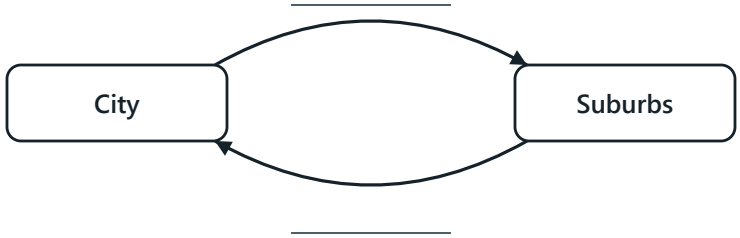
In the system of Problem 12, say in words what each of the four terms $0.9\,b_{n-1}$, $0.3\,r_{n-1}$, $0.1\,b_{n-1}$ and $0.7\,r_{n-1}$ counts.

Problem 15 FREE RESPONSE Draw the diagram for a system

c_n and s_n are the populations of the City and the Suburbs in year n:

$$c_n = 0.8\,c_{n-1} + 0.05\,s_{n-1}, \qquad s_n = 0.2\,c_{n-1} + 0.95\,s_{n-1}.$$

Fill in the moving rates on the diagram.



5. Spreadsheet formulas

Problem 16 FREE RESPONSE The formula, with dollar signs

The spreadsheet below runs the Bayview–Ridgeton model of Problem 12, starting from year 0. (a) What formula goes in cell B6 (Bayview, year 1) so that it can be copied down to later years?

b. What formula goes in C6 (Ridgeton, year 1)?

c. What numbers do B6 and C6 show?

C6 *fx*

	A	B	C	D
1	rate	0.1	Bayview → Ridgeton	
2	rate	0.3	Ridgeton → Bayview	
3				
4	Year	Bayview	Ridgeton	
5	0	600	400	
6	1			
7	2			
8	3			

Problem 17 FREE RESPONSE Copying down without dollar signs

Someone types `=B1*B5+(1-B2)*C5` in C6 of the spreadsheet in Problem 16 (no dollar signs) and copies it down to C7. (a) What formula is in C7?

b. Which cells does it use for the rates, and what goes wrong?

Part 2 · Answers and explanations

1. Periods and frequencies, with units

Problem 1 Period ↔ frequency, with the prefix

$$T = \frac{1}{f}, \qquad f = \frac{1}{T}$$

↓ the prefix flips too

$$1/\text{Hz} = \text{s},\ \ 1/\text{kHz} = \text{ms},\ \ 1/\text{MHz} = \mu\text{s},\ \ 1/\text{GHz} = \text{ns}$$

↓ for example

$$T = \frac{1}{40\ \text{kHz}} = \frac{1}{40}\ \text{ms} = 0.025\ \text{ms} = 25\ \mu\text{s}$$

In words. Take one over the number and flip the prefix: one over kilohertz is milliseconds, one over megahertz is microseconds, one over gigahertz is nanoseconds, and the other way round.

Answers: (a) 25 μs, (b) 0.5 $\mu s = 500$ ns, (c) 4 ms, (d) 0.2 ns, (e) 40 kHz, (f) 250 Hz, (g) 2 MHz.

Common mistakes

- Keeping the prefix: $1/(40\ \text{kHz}) = 0.025$ s or 25 ms. One over kilo is milli: $0.025\ \text{ms} = 25\ \mu s$.
- A bare number ($T = 25$): a period needs a time unit, and quizzes take off points for it.
- Putting a 2π in: $T = 1/f$ has no 2π; that belongs to the angular frequency $\omega = 2\pi f$.

Problem 2 The same time, written other ways

a. $25\ \mu s = 0.025\ \text{ms} = 2.5 \times 10^{-5}$ s.

b. $2.5\ \text{MHz} = 2500\ \text{kHz} = 2.5 \times 10^{6}$ Hz.

Each prefix step (s, ms, µs, ns or Hz, kHz, MHz, GHz) is a factor of 1000: a smaller unit needs a bigger number.

Common mistakes

- Moving the decimal point the wrong way: 1 ms is 1000 µs, so 25 µs is a small number of milliseconds (0.025), not 25 000.
- 2.5×10^{-6} s for 25 µs: count the powers again; 1 µs is 10^{-6} s, so 25 µs is $25 \times 10^{-6} = 2.5 \times 10^{-5}$ s.

Problem 3 How many periods fit?

a. 4: $T_{\text{ref}} = 100\ \mu s$ and $T_{\text{sig}} = 25\ \mu s$, and $100/25 = 4$ (or simply $40/10$).

b. 2: eight signal periods are $8 \times 25 = 200\ \mu s$, which is two reference periods.

Common mistakes

- Answering 1/4 in (a): the signal is the faster one, so several of its periods fit in one slow reference period.
- Answering 32 in (b): the reference is slower, so it makes fewer cycles in the same time: divide by 4.

2. Signal × reference on aligned graphs

Problem 4 Sketch signal, reference and product

$$T_{\text{sig}} = \frac{1}{40\ \text{kHz}} = 25\ \mu s, \qquad 4\ \text{periods} = 100\ \mu s$$

↓ the reference is 4 times slower

$$T_{\text{ref}} = \frac{1}{10\ \text{kHz}} = 100\ \mu s$$

Why: one reference cycle fills the same 100 µs window.

↓ multiply sign by sign, every half signal period (12.5 μs)

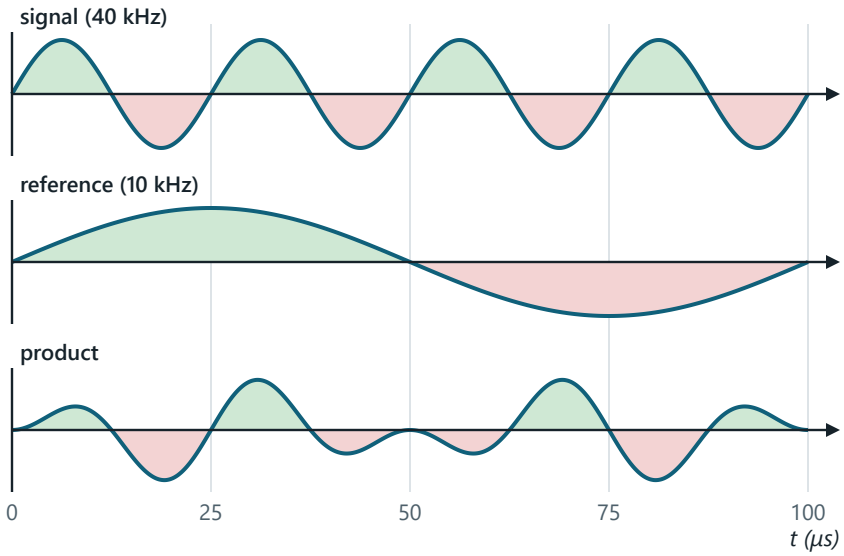
first 50 µs (reference positive): $+, -, +, -$; last 50 µs (reference negative): $-, +, -, +$

↓ the reference is the envelope

Conclusion

the product's humps are small, big, big, small in each half

In words. Label the axis from the period: with 4 grid lines, one signal period per line, 25, 50, 75, 100 µs. The reference makes one slow cycle. Multiply point by point: the product is positive where the two have the same sign, negative where they differ, and its humps are tallest where the reference is near its peak.



Common mistakes

- Labeling the axis in ms (0.1 ms is fine, but 100 ms is not): one over a kilohertz is a millisecond, so $1/(40\ \text{kHz}) = 0.025$ ms.
- Drawing four cycles of the reference too: in the same 100 µs, the 10 kHz reference makes only one.
- Drawing the sum, or a product that is never negative: multiply sign by sign; a negative times a positive is negative.
- Humps all the same height: the product is small where the reference is small.

Problem 5 Pick the product

C. Multiplying point by point: where both are positive or both negative, the product is positive; where one is negative, it is negative. The product is near zero wherever the reference is near zero, so its humps grow and shrink with the reference.

- If you chose **A**: that is the sum of the two, not the product.
- If you chose **B**: a product is negative wherever the two have opposite signs; this one never is.
- If you chose **D**: the signs are right, but the humps should shrink where the reference is near zero.

Problem 6 Label the time axis

Count 5 cycles. One period is $1/(25\ \text{kHz}) = 0.04\ \text{ms} = 40\ \mu s$, so the end of the axis is $5 \times 40\ \mu s = 200\ \mu s$ (0.2 ms).

Common mistakes

- Using one period (40 µs) for the whole axis: the graph shows five.
- 200 ms: check the prefix; one over a kilohertz is a millisecond, so one period is 0.04 ms.

3. Orthogonality

Problem 7 What does orthogonal mean?

Two functions are orthogonal when you multiply them, integrate the product (add up the area under the product curve), and get **zero**:

$$\int x(t)\, y(t)\, dt = 0.$$

On a graph of the product, the area above the axis equals the area below it, so they cancel.

Common mistakes

- "They are perpendicular" or "they have nothing in common" alone: say how you check it, multiply and integrate, and get zero.
- "The product is zero": the product swings above and below zero; it is the total area that is zero.

Problem 8 The integral of the product in Problem 4

first 50 µs: +small, −big, +big, −small

↓ second 50 µs: the same humps with the signs flipped

every positive hump has a negative hump of the same size

↓ the areas cancel

Conclusion

$$\int_0^{100\ \mu s} (\text{signal})(\text{reference})\,dt = 0, \text{ so they are orthogonal}$$

In words. The integral is the area under the product. Its positive and negative humps are equal in size, so the area is zero, and a zero integral of the product is what orthogonal means.

Check with a trig identity: $\sin(a)\sin(b) = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$ turns the product into 30 kHz and 50 kHz cosines, each with whole cycles in 100 µs, so each adds up to zero.

Common mistakes

- "Zero because the product is zero": the product is not zero; its positive and negative parts cancel.
- "Orthogonal" with no reason: link it to the integral being zero.

Problem 9 Which product adds up to zero?

C. Its humps above and below the axis are equal in total, so they cancel: it is the product of two different frequencies (here 1 and 4 cycles).

- If you chose **A**: same frequency: a sine squared is never negative, so its area is positive.
- If you chose **B**: it does dip below the axis, but the humps above are bigger: the total is positive.
- If you chose **D**: all its humps are below the axis: the total is negative, not zero.

Problem 10 Which pairs are orthogonal?

(a) and (c): yes. Different whole-number frequencies, so the product's positive and negative humps cancel over whole cycles.

(b): no. The same frequency, so the product is $4\sin^2(2\pi \cdot 3t)$, never negative; its integral is $4 \times \frac{1}{2} = 2$.

(d): no. The product is $-\cos^2(2\pi \cdot 2t)$, never positive; its integral is $-\frac{1}{2}$, not zero.

Common mistakes

- Calling (b) orthogonal because of the factor 4: a number in front does not change whether the integral is zero.
- Calling (d) orthogonal because of the minus sign: a negative integral is still not zero.

Problem 11 The integral, with units

$$(2\text{ V})(5\text{ V})\sin^2(2\pi \cdot 10\text{ kHz} \cdot t)$$

↓ the average of $\sin^2$ over whole cycles is $\frac{1}{2}$

$$\int_0^T (\text{signal})(\text{reference})\,dt = 2 \times 5\text{ V}^2 \times \tfrac{1}{2} \times 300\ \mu\text{s} = 1500\text{ V}^2\,\mu\text{s}$$

Why: 300 µs holds 3 whole cycles of 10 kHz (and 9 of 30 kHz).

In words. (a) Same frequency: the product never goes negative, so the integral is the average, $A_1A_2/2$, times the time: $1500\text{ V}^2\,\mu s$ ($1.5\text{ V}^2\,\text{ms}$). (b) Different frequencies over whole cycles: zero.

Common mistakes

- $3000\text{ V}^2\,\mu s$: the average of $\sin^2$ is one half, not one.
- Units of V·µs or V²: volts times volts is V², and integrating over time multiplies by the time.

4. Equations from a diagram

Problem 12 Write the system from the diagram

$0.9\,b_{n-1}$ ↓ **stay** in Bayview $(1 - 0.1)$ + $0.3\,r_{n-1}$ ↓ **arrive** from Ridgeton

↓ add them

$$b_n = 0.9\,b_{n-1} + 0.3\,r_{n-1}$$
$$r_n = 0.1\,b_{n-1} + 0.7\,r_{n-1}$$

In words. Each new population is the people who stay (one minus the rate leaving) plus the people who arrive from the other place (that place's rate times its population last year).

Common mistakes

- Swapping the rates: 10% leaves Bayview (the arrow points from Bayview), and 30% leaves Ridgeton.
- Using the rate for the people who stay ($0.1\,b_{n-1}$ in the Bayview equation): those who stay are $1 - 0.1 = 0.9$.
- Leaving out the staying term ($b_n = 0.3\,r_{n-1} - 0.1\,b_{n-1}$): that is only the change.

Problem 13 One year later

$b_1 = 0.9 \times 600 + 0.3 \times 400 = 540 + 120 = 660$ and $r_1 = 0.1 \times 600 + 0.7 \times 400 = 60 + 280 = 340$.

Check: $660 + 340 = 1000$, the same total as year 0, because people only move between the two places.

Common mistakes

- $b_1 = 600 - 60 = 540$: the 120 people who arrive from Ridgeton are missing.
- A total that changes: if $b_1 + r_1 \neq 1000$, a term is missing or a rate is on the wrong population.

Problem 14 What each term means

$0.9\,b_{n-1}$: the people who stay in Bayview. $0.3\,r_{n-1}$: the people who move from Ridgeton to Bayview. $0.1\,b_{n-1}$: the people who move from Bayview to Ridgeton. $0.7\,r_{n-1}$: the people who stay in Ridgeton.

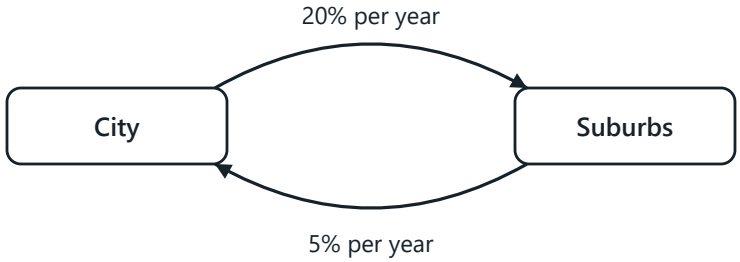
A term in the Bayview equation that multiplies r_{n-1} counts people who were in Ridgeton last year and are in Bayview now.

Common mistakes

- Calling $0.3\,r_{n-1}$ "people who leave Bayview": it multiplies Ridgeton's population, so it counts people coming from Ridgeton.

Problem 15 Draw the diagram for a system

The City keeps 0.8 of its people, so $1 - 0.8 = 0.2$: **20% per year** move from the City to the Suburbs. The $0.05\,s_{n-1}$ in the City's equation counts people arriving from the Suburbs: **5% per year** move from the Suburbs to the City.



Common mistakes

- Labeling the arrows 80% and 95%: those are the fractions that stay, not the ones that move.
- Putting 5% on the arrow out of the City: the 0.05 multiplies the Suburbs' population, so it is a rate leaving the Suburbs.

5. Spreadsheet formulas

Problem 16 The formula, with dollar signs

a. `=(1-$B$1)*B5+$B$2*C5`

b. `=$B$1*B5+(1-$B$2)*C5`

c. 660 and 340, as in Problem 13.

The rates live in B1 and B2, so they get dollar signs: copied down, they stay put. The populations B5 and C5 have none: copied down one row, they become B6 and C6, last year's populations for year 2.

C6 *fx* `=$B$1*B5+(1-$B$2)*C5`

	A	B	C	D
1	rate	0.1	Bayview → Ridgeton	
2	rate	0.3	Ridgeton → Bayview	
3				
4	Year	Bayview	Ridgeton	
5	0	600	400	
6	1	660	**340**	
7	2			
8	3			

Common mistakes

- `=$B$1*$B$5+(1-$B$2)*$C$5`: right for year 1, but locked populations never move, so every year repeats year 1.
- `=$B1*B5+(1-$B2)*C5`: a dollar sign before the letter locks only the column; copying down still changes the row.
- `=(1-$B$1)*B6+$B$2*C6` in B6: year 1 is built from year 0, the row above, not from its own row.

Problem 17 Copying down without dollar signs

a. `=B2*B6+(1-B3)*C6`: copying down one row adds 1 to every row number without a dollar sign.

b. It uses B2 (0.3, the Ridgeton → Bayview rate) instead of B1, and B3 (empty, so 0) instead of B2. C7 then shows $0.3 \times 660 + 1 \times 340 = 538$ instead of the right $0.1 \times 660 + 0.7 \times 340 = 66 + 238 = 304$. Dollar signs on the rate cells, `$B$1` and `$B$2`, keep them fixed.

Common mistakes

- Thinking C7 is the same formula: references without dollar signs move when copied.
- Putting dollar signs on everything: then C7 still uses B5 and C5 and repeats year 1.